

Go Global Travel – R&D Exam

The following exam contains 3 parts.

Read the questions carefully – and do your best to provide clean, well-structured, and efficient code where applicable.

You have 3 days to upload your solution to GitHub and submit the link (with access permissions) to following emails:

HR@goglobal.travel

Part 1 – C# Code

1.

Please describe what issue you can see in IncludeInFinalResults method and how you would refactor this method to solve the issue.

```
public enum RoomAvailableOption
{
    All,
    AvailableOnly,
    NotAvailableOnly
}

public class RoomSearchFilter
{
    RoomAvailableOption enforcedOption;

    public RoomSearchFilter(RoomAvailableOption enforcedOption)
    {
        this.enforcedOption = enforcedOption;
    }
}
```

```

public bool IncludeInFinalResults(Room room)
{
    bool isAvailable = CheckIfRoomAvailable(room);
    switch (enforcedOption)
    {
        case enforcedOption.All: return true;
        case enforcedOption.AvailableOnly: return isAvailable;
        case enforcedOption.NotAvailableOnly: return isAvailable;
        default: return true;
    }
}
}

```

2. Given the following scenario, design a class structure using OOP principles:

- You are developing a ride-sharing application.

You need classes for `Driver`, `Passenger`, `Ride`, and `Vehicle`.

Each ride should be associated with a driver and a passenger.

Ensure proper encapsulation and use of inheritance/polymorphism where applicable.

3. Which design pattern would you use in the following scenarios, and why?

- a) Ensuring only one instance of a logging service exists throughout the application.
- b) Decoupling the creation of complex objects from their representation.
- c) Allowing new functionalities to be added to an object dynamically without modifying its structure.
- d) Providing a simplified interface to a complex subsystem.

4. Explain the importance of SOLID principles and provide an example of a violation of the Open/Closed Principle. How would you fix it?

5. Write a function to find the longest substring without repeating characters.

6. What are the key differences between SQL and NoSQL databases? When would you choose NoSQL over SQL?

Part 2 – SQL

1. Write a query that combines the tables AMOUNTS and ACCOUNTS And returns Amount and Currency For each bank (**BANK**), branch (**BRANCH_ID**) and account (**ACCOUNT_NUM**).

ACCOUNTS:

BANK_ID (PK)	BRANCH_ID (PK)	ACCOUNT_NUM (PK)	CURRENCY
20	621	1001	ILS
20	623	1002	USD
20	90	1003	GBP

AMOUNTS:

ACCOUNT_REC (PK)	AMOUNT
20 621 1001	10000
20 623 1002	20000
20 90 1003	30000

Expected result:

BANK_ID	BRANCH_ID	ACCOUNT_NUM	CURRENCY	AMOUNT
20	621	1001	ILS	10000
20	623	1002	USD	20000
20	90	1003	GBP	30000

2. Consider the following database schema:

```
CREATE TABLE Orders (  
  order_id INT PRIMARY KEY,  
  customer_id INT,  
  order_date DATE,  
  amount DECIMAL(10,2));  
  
CREATE TABLE Customers (  
  customer_id INT PRIMARY KEY,
```

name VARCHAR(255));

3. Write an SQL query to retrieve the top 3 customers with the highest total order amounts in the last 6 months.
4. Write an SQL query to find customers who have placed at least one order every month in the last year.

Part 3 – API

You will create a GitHub repositories search page using their API.

Please choose the from the following stack sets:

- Frontend: Html And JS, Backend: NET 4.72
- Frontend: React, Backend: NET 8

1. The user will type the repository he would like to search.
2. When searching (by pressing a button or using the Enter key) you will perform a request to:
`https://api.github.com/search/repositories?q=YOUR_SEARCH_KEYWORD`
3. Render the results as gallery items where each item will show to repository name, avatar of the owner and a bookmark button and sendEmail button.
4. When a user bookmarks a repository you will store the entire result to the user's session (**Use ASP.NET session for .Net 4.72 or other storage for .Net 8**).
5. When a user click sendEmail button a popup with email address input and send button will appear.
6. After send clicked – email with selected repository basic details will be sent to the email address
7. (Bonus) - Add a Bookmark screen that will show all the bookmarked repositories with option to unbookmark result.

Make sure you leave a Readme.md file so we can how to run the project.